

Pauli Correlation Encoding for Distribution Grid Expansion Planning

Thousands of build-or-no-build line decisions on tens of qubits, three measurement settings at any size, and a hardness ladder to benchmark them on

Team eQoSystem (AiQC, Ai and Quantum Community)

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<https://github.com/AshrafBoussahi/eQoSystem-EON-GridExpansion>

Problem statement: **E.ON**, *Quantum-Enabled Grid Expansion Planning for Distribution System Energy Networks*

Summary. E.ON’s problem is combinatorial along one axis: the number of build-or-no-build line decisions. That is exactly the axis along which the standard one-qubit-per-variable quantum mapping fails, and exactly the axis along which Pauli correlation encoding (PCE) is cheapest. PCE stores each binary decision in the sign of a k -body Pauli correlation, so an n -qubit register carries up to $3\binom{n}{k}$ decisions and, because the strings split into three commuting families, the device runs three measurement settings whatever the problem size. We have already built and published the full workflow on transmission-planning problems with real power-flow coefficients (45 decisions on 6 qubits, 105 on 7, 9,009 on 15), measured its three-setting property on a 108-qubit superconducting processor, shown that discrete native-gate PCE circuits beat variational PCE at low quantum budget up to 858 variables on 13 qubits, and won first place globally in the Rigetti advanced track of the Q-Volution hackathon with a PCE solution to a 180-node grid optimisation problem. This proposal re-targets that stack at E.ON’s line-addition formulation, adds a QOBLIB-style hardness ladder of distribution expansion instances, and commits to a Qiskit-executable, reproducible PoC with numeric success criteria.

1. Problem Framing

Our reading of the problem. A distribution system operator holds a network $G = (V, E_0)$, a set of candidate additions E_c (new ties, parallel conductors, conductor upgrades on existing corridors), and a scenario set S of future operating conditions with probabilities p_s . The decision is $x \in \{0, 1\}^m$, $m = |E_c|$: which candidates to build. The objective E.ON names is congestion, the amount by which line flows exceed ratings, summed over lines and scenarios, traded against investment cost under a budget (a two-objective structure of the kind treated in [6]). We adopt the binary linear programming family of reference [4] of the challenge statement, in which candidate lines are scored by their influence on congested branches, and we keep the flow physics linear as in the demand coincidence models of reference [1]:

$$\min_{x \in \{0,1\}^m} \sum_{s \in S} p_s \sum_{l \in E_0 \cup E_c(x)} \max(0, |f_l^s(x)| - F_l) + \lambda \sum_{e \in E_c} \kappa_e x_e \quad \text{s.t.} \quad \sum_e \kappa_e x_e \leq K, \quad x \in \mathcal{T}, \quad (1)$$

where $f_l^s(x)$ is the linearised flow on line l in scenario s given the built set, F_l the thermal rating, κ_e the cost of candidate e , K the budget, and $\mathcal{T}$ the topology rules a DSO imposes (radial operation of low-voltage feeders, at most one upgrade tier per corridor). Outputs are the ones the statement asks for: the set of lines to build, the objective value, and the congestion before and after. The formulation is deliberately a variant of the network expansion literature rather than a new invention; the challenge asks for hard instances of such a variant, and that is where our contribution starts.

Why the decision count is the barrier. A medium-voltage subgrid with a few hundred corridors and three upgrade options per corridor is already $m \approx 10^3$. The mainstream quantum optimisation proposals (QAOA, VQE on an Ising Hamiltonian, annealing) spend one qubit per binary variable, so $m = 10^3$ needs a thousand-qubit register at a fidelity no device offers today. The gap is a property of the encoding, not of the hardware roadmap.

Why PCE gives a credible route. Pauli correlation encoding [7] reads decision i from the sign of the expectation value of a k -body Pauli string Π_i on an n -qubit state, $x_i = \frac{1}{2}(1 + \text{sign}(\langle \Pi_i \rangle))$. Taking strings from the three uniform families $X^{\otimes k}$, $Y^{\otimes k}$ and $Z^{\otimes k}$ over qubit subsets gives capacity $m_{\max} = 3\binom{n}{k}$: 45 decisions on 6 qubits, 105 on 7, 2,000 candidate lines on 18 qubits at $k = 3$ (capacity 2,448), and 4,500 on 14 qubits at $k = 5$. Strings within a family commute, so one basis rotation reads a whole family at once and the number of measurement settings is three, independent of m and n . Both properties have been measured by us, not argued: Table 1 lists what already runs.

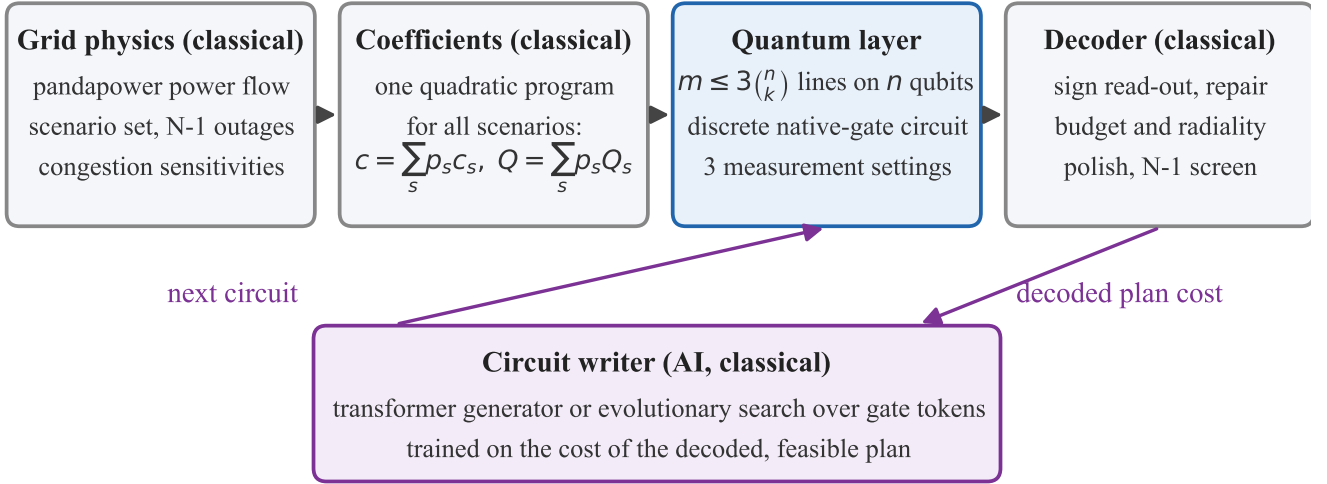
What a successful outcome would change for E.ON. Today candidate reinforcements are screened by rules and experience and iterated through simulation, which is why plans tend to be conservative. A solver whose device workload stays at three circuits while the candidate set grows into the thousands lets a planner evaluate the joint choice over the whole candidate set, across all scenarios, and return a ranked set of lines with the congestion relief each delivers. We claim no quantum speed-up at the scale where a mixed-integer solver certifies optima in milliseconds; we claim a resource profile that is bounded along the axis on which the problem explodes, and we propose to test it on instances certified to be hard.

2. Technical Approach

Paradigm. Gate-based, hybrid classical-quantum, and non-variational: the quantum circuit holds no trainable angle. A classical circuit writer proposes a discrete native-gate circuit, the device prepares the state and is measured in three settings, a classical decoder turns the correlation signs into a feasible plan, and the plan’s exact objective is the training signal for the writer. Figure 1 shows the loop; each stage below exists in code today.

Table 1: What already runs, end to end, before the PoC starts. All numbers are from our published result files (repository in the header). Rows 1 to 4 are storage-siting and graph problems with the same quadratic form as Eq. (1); row 5 is the winning Q-Volution run.

Problem	m	n	k	Where and how it ran	Settings
Storage siting, IEEE-14/118	45	6	2	exact simulation; Rigetti Cepheus-1-108Q, 3 jobs, 3,072 shots	3
Storage siting, IEEE-118	105	7	3	exact simulation; Rigetti Cepheus-1-108Q, 3 jobs, 3,072 shots	3
Weighted Max-Cut, 3-regular and ER graphs	858	13	3	exact simulation, discrete search against variational PCE	3
Graph partitioning, largest executed	9,009	15	5	exact simulation, 30.4 h on one 64-core instance	3
Grid Max-Cut, real US topology (180 nodes)	180	12	2	Rigetti Ankaa-3 QPU, first place, Q-Volution 2026	3



Output: lines to build, congestion before and after, cost

Figure 1: The hybrid loop. Everything that grows with the number of scenarios or with the physics stays classical; the quantum layer sees one quadratic program and returns m correlation signs from three measurement settings.

2.1 From grid physics to one quadratic program

Flows in Eq. (1) are linear in injections for a fixed topology, and the effect of adding candidate e on the flow of line l is a line-outage-style sensitivity (the influence score of reference [4]). Expanding the congestion term to second order in x turns each scenario into a quadratic binary program $c_s^\top x + x^\top Q_s x$, and because the build decision is taken before the scenario is revealed, the expected objective is the single program with $c = \sum_s p_s c_s$, $Q = \sum_s p_s Q_s$. That aggregation is exact and classical, so *scenario count costs nothing at the quantum layer*. We have used this construction with thirteen weather scenarios clustered from 105,408 five-minute intervals of the RTS-GMLC dataset: the scenario set changed the decision on 77.5% of 40 instances and was strictly better under the high-stress scenario on 77.5% and worse on none, at identical qubits, depth and shots. Coefficients came from a DC optimal power flow whose locational price decomposition we verified numerically to 6.4×10^{-4} \$/MWh. For E.ON’s medium- and low-voltage networks the same builder will read pandapower models [10] (IEEE and CIGRE cases, SimBench feeders [11]) and use the linear DistFlow branch model [12] on radial feeders, so voltage limits enter the congestion term alongside thermal ratings.

2.2 The encoding, and what it buys

Each candidate line is assigned to one k -body Pauli string on the register. Upgrade tiers on a corridor (for example three conductor sizes) are a monotone chain of bits, a domain-wall encoding [9], chosen because an unconstrained bit string is already a valid chain 97.5% of the time against 13.8% for plain binary and 0% for one-hot, and one bit flip moves the tier by exactly one step. Table 1 gives the measured compression: 7.5 decisions per qubit at $m = 45$, 15 at $m = 105$, 600 at $m = 9,009$. Over that range the decision count grew 35.8-fold, the register 1.7-fold, and the measurement cost not at all: three settings and $3 \times 1,024$ shots per circuit evaluation at every size.

2.3 Circuits written, not optimised

The circuit is a sequence of tokens from a fixed vocabulary: single-qubit rotations on a fixed angle grid, and either two-qubit entangling tokens or a fixed brickwork of controlled-Z gates matching device connectivity. Two circuit writers exist and both

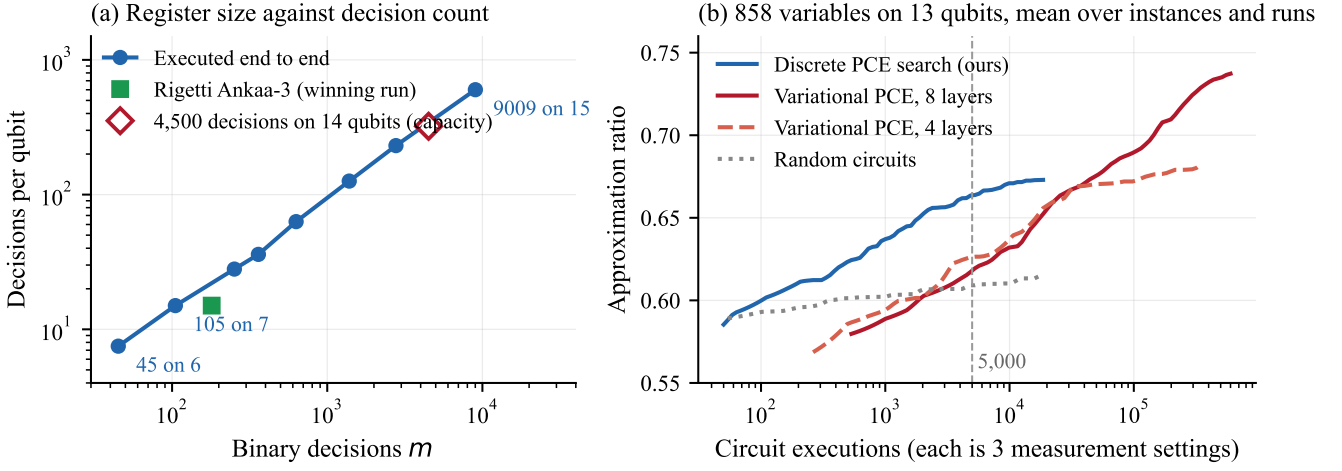


Figure 2: (a) Decisions per qubit across every configuration executed end to end, the winning Ankaa-3 run, and the register a 4,500-decision planning instance (500 corridors at nine upgrade tiers) would occupy. (b) Solution quality against quantum budget at 858 variables on 13 qubits, mean over instances and runs: the discrete PCE search (no circuit parameters, no gradients) leads below 5,000 executions; the variational method overtakes only after an order of magnitude more.

are gradient-free on the device:

- a decoder-only transformer in the style of the generative quantum eigensolver [8], trained with a listwise preference loss on the cost of the *decoded, feasible* plan. Its one governing design quantity is the gate budget: raising it from 19 to 59 gates lifted the share of 80 grid instances solved to the certified optimum from 23.8% to 48.8% (paired Wilcoxon, better on 45, worse on 4, $p = 2.4 \times 10^{-7}$);
- an elitist evolutionary search over the same token space, used at 60, 252 and 858 variables. Below 5,000 circuit executions it leads variational PCE with parameter-shift gradients by 4 to 9 points of approximation ratio at every size (Figure 2b); the variational method overtakes only above 10^4 to 10^5 executions and needs 15 to 30 times the budget for its final margin. The circuits contain no continuous angle, need no compilation, and run unchanged on hardware.

The reward combines the exact decoded objective with the smoothed tanh surrogate of [7]. We found that the surrogate is what makes decoded signs survive finite shots: at 252 variables, circuits trained on the exact objective alone kept an approximation ratio of 0.591 when read from 1,000 shots, against 0.698 with the smoothed term, because the latter drives correlator magnitudes from 0.019 to 0.109. This is the design rule for the hardware campaign in Section 5.

2.4 Decoder: a feasible plan every time

Sign read-out gives a bit string, not a plan. The decoder repairs each tier chain, enforces the budget by removing the least valuable additions, checks the topology rules, then polishes with a joint single- and pairwise-flip search and a small integer program over the touched variables. Infeasible output is structurally impossible: whatever the device returns, the planner receives a buildable set of lines. Every decoded plan is then screened classically against the full N-1 single-outage set with a security-constrained power flow that is allowed to shed load, so the resilience effect of a plan is a measured difference on identical outages. On IEEE-57 with 200 MW of added load, the sited plan raised the share of outages served without shedding from 36.4% to 96.1% and cut expected unserved energy by 93.5%.

2.5 Qiskit compatibility layer, already in place

Our PCE circuit-synthesis code base builds every circuit as a Qiskit `QuantumCircuit`; the same three X/Y/Z measurement circuits run in the exact statevector path and in Aer with and without noise models, and are submitted unchanged to IBM hardware through the Qiskit Runtime path of the library. Read-out is a Walsh-Hadamard transform of the three count tables that returns all m correlators at once, in $O(n^2)$ per basis and independent of m , and it agrees with Qiskit’s Pauli expectation values to 10^{-10} . The grid package emits OpenQASM 3, which Qiskit loads directly, and that is the path we used on Rigetti devices through qBraid. E.ON can therefore execute and validate every result in Qiskit on day one of the PoC, not at its end.

3. Feasibility and Resource Requirements

Software. Two public code bases exist and are the basis of the PoC. qGridX (<https://pypi.org/project/qgridx/>; <https://github.com/ashrafboussahi/qGridX>) is a pip-installable package of 69 modules covering the power-flow layer, the encoding, the transformer writer, the decoder, N-1 screening, statistics, and the hardware path, with every published number reproduced by a one-line command and two qBraid notebooks. genpce (6,758 lines, 26 unit tests) is the Qiskit-native PCE circuit-synthesis library with the evolutionary writer, the Walsh-Hadamard read-out, the variational baseline and the IBM Runtime path. The PoC work is integration and a new coefficient builder, not new infrastructure.

Data. Open only: pandapower IEEE and CIGRE medium- and low-voltage cases, SimBench feeders, IEEE 33-, 69- and 123-bus radial systems, RTS-GMLC time series for scenarios, and synthetic feeders generated to match their graph statistics.

Table 2: Work plan for the PoC sprint. Every deliverable is a file in the public repository; WP1 is the challenge’s stated foundation and is released first so that E.ON and other teams can use it.

Work package	Weeks	Deliverable
WP1 Hardness ladder	1 to 4	six instance sizes, $m = 45$ to 2,000, on open MV/LV networks; certified optima or best-known values; HiGHS and Gurobi time-to-optimality and root gap
WP2 Line-addition solver	3 to 8	pandapower coefficient builder, decoder rules for budget, tiers and radiality, simulation campaign to 15 qubits with all baselines and paired statistics
WP3 Hardware campaign	7 to 12	depth sweep, Bell controls, three-setting runs at $m = 45$ and $m = 105$ on Rigetti and IBM with mitigation; raw counts and analysis notebooks
WP4 Scale and simulator study	9 to 14	24- to 30-qubit runs, MPS bond-dimension sweep, E.ON anonymised subgrids on the same builder, Qiskit validation package
WP5 Report and paper	13 to 16	reproducibility bundle with one command per figure, final report, joint paper draft with E.ON

E.ON’s anonymised subgrids will be used when released, on the same builder.

Compute. Exact simulation to 15 qubits runs on a laptop; the 9,009-decision instance took 30.4 hours on one 64-core cloud instance, which is the largest exact run we will need. The 2,000-line rung fits 18 qubits at $k = 3$ and is exactly simulable. For the simulator study we deliberately use $k = 2$ on 24 to 30 qubits (828 to 1,305 line decisions): two-body correlators are the largest and easiest to read on hardware, and a register above 24 qubits is where the question of Section 5 becomes meaningful; 30 qubits is the top of exact statevector on one 64-core instance, and Aer’s matrix-product-state backend covers the rest. qBraid compute is already provisioned.

Hardware. Rigetti through qBraid (we have run Ankaa-3 and Cepheus-1-108Q) and IBM through Qiskit Runtime. Our measured costs set the plan: one 1,024-shot job returns one measurement setting, so one circuit evaluation is three jobs, and a 2,000-evaluation campaign is 6,000 short jobs, submitted in batches. On Cepheus the $m = 45$ and $m = 105$ circuits routed to 32 and 41 two-qubit gates at depth 58 and 81.

Assumptions and constraints, stated. (i) Flows are linearised (DC for meshed medium voltage, linear DistFlow for radial low voltage); the decoder re-evaluates every plan with a full pandapower power flow, so the linearisation affects the search surface, not the reported objective. (ii) Scenarios are a finite weighted set. (iii) Correlator magnitudes must clear the shot-noise floor $1/\sqrt{N} \approx 0.03$ to be read reliably, which sets a depth bar that our device campaigns have located from both sides: two-qubit Bell correlators on eight Cepheus edges came back at mean magnitudes 0.74 to 0.82 with 24 of 24 signs correct, while 59-gate generated circuits (32 two-qubit gates after routing) fell below the floor. The hardware campaign therefore opens with a depth sweep on the day’s calibration, trains generated circuits under the gate budget it locates, uses the smoothed term of Section 2.3, and reads with measurement-error mitigation [14]. (iv) Exact simulation beyond about 30 qubits is out of scope; above it the PoC uses hardware and bounded-bond-dimension MPS only.

Timeline. Table 2 lays out sixteen weeks in five overlapping work packages, each with a deliverable that exists as a file.

4. Expected Impact

A successful PoC demonstrates five things, each with a number attached.

1. **A hardness ladder for distribution expansion**, in the sense of QOBLIB [5]: at least six instance sizes from $m = 45$ to $m = 2,000$ candidate lines on open networks, each with a certified optimum or best-known value, time-to-optimality for HiGHS and Gurobi, and root-gap, and at least one rung at $m \leq 500$ where a commercial solver does not prove optimality within one hour. This is the challenge’s stated foundation and it is reusable by every other team and by E.ON.
2. **A quantum workflow whose device workload does not grow with the candidate set**: three measurement settings and a fixed shot budget from $m = 45$ to $m = 2,000$, with the register between 6 and 30 qubits.
3. **Solution quality on the ladder.** Targets: the certified optimum on at least 50% of $m \leq 105$ instances at a matched 10,000-evaluation budget (we reach 48.8% on our transmission family today), and at least 0.95 of the best-known value on $m \geq 500$ instances within 20,000 evaluations.
4. **Hardware execution of at least one hard instance** at $m = 45$ and $m = 105$ on a superconducting processor, with sign agreement against exact simulation above 90% after mitigation, and a decoded plan whose congestion relief is within 10% of the simulated one.
5. **A Qiskit-executable deliverable** that E.ON runs end to end in under three hours per instance.

For the use case, the change is from screening a handful of rule-based reinforcement candidates to evaluating the joint choice over hundreds to thousands of candidates across the full scenario set, with every plan N-1-screened before it reaches a planner. E.ON’s closing wish, a joint scientific paper, is one we share: the ladder, the encoding and the device measurements are each publishable on their own.

Table 3: Success criteria for the PoC. Each is measured by the protocol above and reported whether or not it is met.

Criterion	Target
Hardness ladder released with certified values and solver times	6 rungs; one rung at $m \leq 500$ unproven by Gurobi in 1 h
Device workload across the ladder	3 settings, fixed shots, 6 to 30 qubits at every rung
Exact-match rate on $m \leq 105$ at 10,000 evaluations	$\geq 50\%$ of instances (48.8% today on the transmission family)
Ratio to best-known on $m \geq 500$ at 20,000 evaluations	≥ 0.95 , paired against SA and tabu at the same budget
Hardware, $m = 45$ and $m = 105$	sign agreement $\geq 90\%$ after mitigation; decoded relief within 10% of simulation
Simulator study, 24 to 30 qubits	bond dimension at which the MPS objective departs from exact, with CI
Qiskit execution by E.ON	end to end under 3 h per instance from the public package

5. Validation Plan

Unit of analysis and statistics. The unit is the instance, never the run. Every comparison is paired on identical instances at a matched evaluation budget and reported with a paired Wilcoxon signed-rank test for objective values, McNemar’s exact test for exact-match rates, and Wilson intervals for proportions; at least 20 instances per rung. This is the protocol behind every number quoted above.

Baselines, all at matched budget. Exact MIP (HiGHS, Gurobi academic licence) with time-to-optimality; simulated annealing; tabu search; multi-start greedy; random circuits through the same decoder (the ceiling the decoder can extract from arbitrary input); variational PCE with parameter-shift gradients; and one-qubit-per-variable QAOA on an extra $m \leq 30$ rung added for that comparison, the only size at which it fits on hardware. On our current family the decoder-matched control ties our solver ($p = 0.402$) and 20-restart greedy loses 62 instances to 2 ($p = 5.7 \times 10^{-12}$); we will publish the same table for the new ladder, including the rungs where classical metaheuristics win.

Metrics. Congestion before and after (MW above rating, summed over lines and scenarios, and the count of violated lines); investment cost; approximation ratio to the certified or best-known value; share of instances solved to the optimum; N-1 secure fraction and expected unserved energy of the decoded plan; end-to-end wall clock.

Hardware protocol. Offline verification first: the emitted circuit is executed two ways (package simulator and interpreted QASM) and must agree to numerical precision, as it did before the Cepheus campaign (fidelity 1.000000000000, correlators to 3×10^{-12}). Then a Bell control on every used edge, then the depth sweep, then the three settings per circuit with read-out calibration in the same session. Reported quantities: sign agreement and magnitude retention against exact simulation, both bit orderings, and the decoded objective. Success is defined above; failure modes are reported with the same tables.

Secondary objective: simulator performance. At $n \leq 20$ every circuit is exactly simulable, so the MPS question is only meaningful on the 24- to 30-qubit rung. There we execute the trained circuits on Aer’s matrix-product-state backend at bond dimensions $\chi \in \{2, 4, 8, 16, 32, 64\}$ and report the decoded congestion objective against the exact or hardware value. Our trained PCE states are strongly entangled (half-chain entropy 2.1 of 3 bits at six qubits, all correlators between 0.13 and 0.37), so a bounded- χ simulator is expected to degrade; the deliverable is the measured χ at which it does, with confidence intervals.

Reproducibility. Seeds, CSV result files, one command per figure, and qBraid notebooks, as in the two repositories today. Table 3 collects the criteria that define success.

6. Hybrid / Cross-Domain Integration

The split in Figure 1 is deliberate. Power-flow physics, scenario aggregation and N-1 screening stay classical because they are linear algebra that scales with grid size and scenario count, and none of it belongs on a device. The AI layer, a transformer or an evolutionary search over gate tokens, stays classical because that is where a learning signal can be accumulated across thousands of proposals at no quantum cost, and because it removes both the ansatz design problem and every quantum gradient. The device does the one thing the classical layers cannot do cheaply: hold m coupled binary decisions as m correlation signs of one entangled state and return them from three measurements. The decoder closes the loop by guaranteeing feasibility, so the quantum output is always a plan a planner can act on. The same pipeline runs on a 64-core cloud instance with the device replaced by a statevector or MPS simulator, which is how every rung of the ladder is validated before device time is spent.

7. Team Capability

Track record on this exact method and this exact domain. eQoSystem, a team of AiQC (Ai and Quantum Community), won first place globally in the Advanced Rigetti Computing Challenge at the Q-Volution hackathon hosted by Girls in Quantum (March 2026), after being selected before the event for a QPU access grant. The task was a 180-node weighted Max-Cut instance derived from a real US power-grid topology; our PCE solution mapped the whole grid onto 12 qubits without partitioning, ran on the Rigetti Ankaa-3 QPU, and returned a cut above the classical baseline. The Rigetti judges credited the team with “a brilliant, deep understanding of the problem” (<https://www.aiqcommunity.org/news/q-volution-first-place>;

solution at https://github.com/AshrafBoussahi/Rigetti_Quantum_Grid_Optimization_eQoSystem). The team also represented Algeria and Africa at MIT's iQuHACK 2026.

Since then. We built and published qGridX on PyPI, ran two device campaigns on the Rigetti Cepheus-1-108Q through qBraid, wrote the PCE circuit-synthesis library with its Qiskit Runtime path, and produced the results in Table 1. Everything in Sections 2 to 5 is a description of code we run, not of code we plan.

Members.

- **Achraf Boussahi** (lead). Co-director and Chief Scientific Officer, AiQC; research intern, Constantine Quantum Technologies (CQTech); ESI Sidi Bel Abbes. Quantum optimisation, PCE, hardware campaigns on Rigetti Ankaa-3 and Cepheus-1-108Q, author of qGridX and of the DC-OPF and N-1 layers.
- **Abir Chekroun**. Co-director and Head of Quantum Projects, AiQC; ESI Sidi Bel Abbes. Quantum algorithms and PCE pipelines; Q-Volution winning team; quantum simulation packages.
- **Zakaria Loughi**. Co-director and Principal AI Engineer, AiQC; ESI Sidi Bel Abbes. Transformer training, machine learning systems and software engineering; Q-Volution winning team.
- **Mouadh Abderrahmane Assal**. Co-director and Head of Advanced AI Research, AiQC. Reinforcement learning and energy forecasting (renewable energy forecasting datathon, quantum-assisted deep RL).
- **Hiba Menacer**. team member; role and affiliation as entered in the portal team profile.

Quantum optimisation, power-system modelling, machine learning and software delivery are each covered by at least one member who has shipped it, and the lead has run every hardware platform the plan relies on.

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